ARPAN BANIK

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EDUCATION

Swami Vivekananda Institute Of Science And Technology 08/2024 – Present
B.Tech in Electronics and Communication Engineering — CGPA: 7.05 | Kolkata, India

Basirhat Govt Polytechnic 10/2021 – 07/2024
Diploma, Electrical Engineering — Percentage: 75% | Basirhat, India

PROJECTS

Near It... — Smart Service Booking & Management Platform | GitHub | Live 09/2026 – Present
Tech Stack: React, Vite, Redux Toolkit, Tailwind CSS, Node.js, Express.js, MongoDB, Mongoose, JWT, Razorpay, Socket.IO, Leaflet, Cloudinary, Nodemailer

- Built a full-stack, three-role (user / provider / admin) service marketplace with REST APIs under /api/v1 and 14+ MongoDB models (bookings, payments, refunds, reviews, notifications) on a layered controller–service–model backend.
- Implemented JWT authentication with HTTP-only cookies and refresh tokens, email OTP verification (Nodemailer + Brevo SMTP), and backend-enforced role-based access control so roles cannot be escalated from the frontend.
- Designed a booking lifecycle (PENDING → ACCEPTED → ON_THE_WAY → STARTED → COMPLETED) with provider accept/reject actions, status-history tracking, and a provider onboarding flow with admin document verification.
- Integrated Razorpay with server-side amount calculation and signature verification, and built a refund workflow with admin processing and duplicate-request protection.
- Developed nearby-provider discovery using MongoDB geospatial queries with Leaflet/OpenStreetMap maps, and Socket.IO for real-time notifications and booking updates; deployed the frontend on Vercel.

Pluto — Full-Stack Social Media Platform (Facebook / Instagram-style Clone) | GitHub | Live 09/2025 – Present
Tech Stack: React, Redux Toolkit, Tailwind CSS, Node.js, Express.js, MongoDB, Socket.io, Cloudinary, JWT

- Built a full-featured Facebook/Instagram-style social media platform with real-time capabilities using the MERN stack and Socket.io for live notifications, likes, and comment count updates.
- Implemented JWT-based authentication, profile management with Cloudinary avatar/cover image upload, and automatic cleanup of old images on update.
- Developed a Reels/Shorts-style vertical video feed with IntersectionObserver-based autoplay, infinite scroll, and view tracking for both posts and videos using TTL-based Mongoose models.
- Built a complete engagement system including post/video likes with optimistic UI updates, nested comments, follow/unfollow, saved posts (Watch Later), and watch history with tabbed UI and skeleton loaders.
- Engineered real-time Socket.io rooms per post/video for live like and comment count sync across all connected clients, with Redux Toolkit slices handling all async state via thunks.

RAG Project — LLM-Powered Application (Python) | GitHub 10/2026 – Present
Tech Stack: Python, Groq API, LLMs, Prompt Engineering, uv, python-dotenv, Git

- Building an LLM-powered Python application on the Groq API and Python SDK, sending chat requests with system / user / assistant roles and handling API responses and errors.
- Experimented with prompt engineering and generation controls (temperature, max_tokens), and tracked token usage and finish_reason to manage output quality and cost.
- Set up an isolated uv virtual environment with secure API-key handling through environment variables, version-controlled on GitHub.
- Extending the project into a Retrieval-Augmented Generation pipeline (embeddings, vector search, document retrieval).

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C, NoSQL
Frontend: HTML, CSS, ReactJS, NextJS, Redux Toolkit, Tailwind CSS, React Router, React Hook Form, Zod
Backend: Node.js, Express.js, Socket.io, REST APIs, JWT Authentication, Razorpay, Nodemailer
Databases: MongoDB, MySQL
AI / LLM: Groq API, Prompt Engineering, LLM API Integration, RAG (learning)
Cloud & Tools: Cloudinary, Vercel, Git, GitHub, CI/CD (GitHub Actions), Leaflet

COURSEWORK / ADDITIONAL SKILLS

Data Structures & Algorithms • Web Development • MERN Stack Development